FastConduct v1.2 — User Manual



Ionic Conductivity Calculator (kMC)

What's New in v1.2

Robust CIF reader. Files written by GSAS2CIF, FullProf, ICSD or VESTA are now read correctly: loop_ tags placed on the same line, data rows of neighbouring loops, several data_ blocks, Cartesian coordinates (_atom_site_Cartn_*), positions exactly on the cell boundary, and a cross-check of the atom count against _chemical_formula_sum. Tested on 28 CIF files from different sources.

CIF export writes real cell angles for hexagonal, monoclinic and triclinic cells (previously alpha/beta/gamma were always 90 degrees), plus _cell_volume. Structures exported for DFT are now geometrically correct.

New Hw convention for guest-water protons - see the dedicated section below.

Save Plot -> PDF works again.

Download size is 57 MB instead of 276 MB (the executable is still a single .exe).

1. Overview

FastConduct is a graphical user interface (GUI) application for calculating ionic conductivity and tracer diffusion coefficients using the **kinetic Monte Carlo (kMC)** method. It is designed for materials scientists, chemists, and physicists working with solid electrolytes, electrode materials, and other ion-conducting systems.

Unlike many existing kMC tools, FastConduct offers two critical advantages:

1. **User-defined migration energies** — you can input migration barriers obtained from high-

accuracy **DFT-NEB** (Density Functional Theory with Nudged Elastic Band) calculations or any other source, rather than being restricted to built-in empirical potentials.

2. **Tunable vacancy concentration** — the vacancy percentage can be freely varied to study its effect on ionic transport, enabling systematic screening across different stoichiometries and defect levels.

Key Features

- Import crystal structures from **POSCAR**, **CONTCAR**, or **CIF** files
- Build supercells with user-defined vacancy concentrations
- Simulate thermally activated ion jumps on the lattice
- Calculate **σ** (ionic conductivity, S/cm) and **D** (diffusion coefficient, cm^2/s)
- Generate **Arrhenius plots** ($\log_{10}\sigma$ vs 1000/T) with linear fitting
- Automatic nearest-neighbor detection and jump-length estimation
- Optional **ion–ion repulsion** model
- Export vacancy structures as **CIF** files
- Save results and plots for publication

2. System Requirements

Parameter	Requirement
OS	Windows 10 / 11 (64-bit)
RAM	Minimum 4 GB (8+ GB recommended for large supercells)
Disk space	~250 MB (57 MB download + unpacking to a temporary folder)
Dependencies	None — standalone executable
Download size	57 MB (single .exe, no installer)

Note: The program is distributed as a single .exe file. No administrator privileges are required; only write access to the program folder is needed.

3. Installation & Launch

3.1. First Launch

1. Copy FastConduct.exe into a dedicated folder (e.g., C:\FastConduct\).
2. *(Optional)* Place a logo image named FastConduct.png or logo.png in the same folder to display it in the header.
3. Double-click FastConduct.exe.
4. A splash screen appears for 2.5 seconds, after which the main window opens.

3.2. Working Directory

On first run, the program creates a subfolder named vacancy_structures/ next to the executable to store output CIF files. Ensure you have write permissions in the program directory.

4. Interface Description

The interface is organized into nine functional sections.

4.1. Load Structure

- **Load File** — opens a file dialog.
- Supported formats: .vasp (POSCAR, CONTCAR), .cif.
- After loading, the chemical formula, elements, lattice parameters, and total atom count are displayed.

The program supports parallel calculations. Clicking "New Window" opens a new application window.

4.2. Supercell Size

- **Nx, Ny, Nz** — integer multipliers of the primitive cell along each axis (≥1).
- **Total** — displays the total number of atoms in the supercell.

4.3. Working Ion

- **Ion** — dropdown list of elements detected in the loaded structure.

- **Ion charge (q)** — formal charge of the ion in units of elementary charge (e.g., 1 for Li^{+}, 2 for Zn^{2+}, -2 for O^{2-}).
- **Vacancy concentration (%)** — percentage of working-ion sites to be vacated (e.g., 5 = 5%).
- The program automatically reports how many atoms will be removed.

4.4. Migration Energy (E_m)

- **E_m (eV)** — the migration barrier. This can be taken from **DFT-NEB calculations**, bond-valence site energy (BVSE) landscapes, or experimental estimates.
- **Enable ion–ion repulsion** — activates a softBV-style repulsion model [1]:
  - **alpha (eV/neighbor)** — repulsion energy per occupied neighbor.
  - **Cutoff (Å)** — interaction radius; auto sets it to 1.5× the nearest-neighbor distance.

Guest-Water Protons: the Hw Convention

If the structure contains a guest water molecule and the diffusing species is a proton - as in MOFs with water in the pores or in hydrated solid polymer electrolytes - the water protons must be separated from the structural hydrogens (C-H, N-H and O-H groups of the framework). FastConduct selects the mobile species by the element symbol only and strips digits from labels (H1, H2 -> H), so the only reliable way to separate the two kinds of hydrogen is a dedicated symbol: Hw.

How to use it:

1. In the CIF file set _atom_site_type_symbol = Hw for the two hydrogens of every water molecule; structural hydrogens keep the symbol H. The helper script mark_water_h.py does this automatically (it treats any oxygen with two hydrogens within 1.25 A as a water molecule) and can convert the file back before a DFT calculation.
2. Load the file in FastConduct - Hw is selected as the working ion automatically.
3. Set Ion charge (q) = +1. The program enters this value automatically when Hw is selected.
4. Set the vacancy concentration. It is applied to the Hw sublattice only, so the framework hydrogens never

enter the jump network.

The results report contains the line 'Species: protons of a guest water molecule (Hw)'. If H is selected while Hw is present, the program warns that structural hydrogens will also be counted.

Important: Hw is not a chemical element. VESTA, ASE and pymatgen will not open such a CIF, and the file must not be used as DFT input. Convert it back with mark_water_h.py --unmark.

4.5. Diffusion Dimensionality

- **1D** — chains / tunnels: D = ⟨MSD⟩ / (2t)
- **2D** — layers / planes: D = ⟨MSD⟩ / (4t)
- **3D** — bulk / volume: D = ⟨MSD⟩ / (6t)

4.6. kMC Simulation Parameters

- **Number of runs** — independent simulation runs (recommended: 5–10) for statistical averaging.
- **Available jumps (N_p)** — number of neighboring sites considered for each ion (auto-detected from the structure).
- **kMC steps per run** — maximum number of Monte Carlo steps (default: 10,000,000).
- **Equilibration steps** — steps performed before data collection to thermalize the system.
- **Output CIF folder** — directory name for saving vacancy structures.

4.7. Temperatures (K)

- Enter up to 5 temperatures separated by commas, e.g.: 250, 275, 300, 325, 350

4.8. Calculation Parameters

- **Attempt frequency ν_0 (Hz)** — typical value: 10^{13} Hz.
- **Max jump length (Å)** — maximum allowed hop distance; 6.5 Å recommended for cations, 4.5 Å for anions.
- **Max simulation time (s)** — physical time limit per temperature.

4.9. Results & Plot

- **Results** text area — full report with conductivity, diffusivity, and statistics. Right-click to copy or save.
- **Arrhenius Plot** — interactive matplotlib figure with linear fit, R^2, and activation energy.
- **Save Plot** — export the figure as PNG, PDF, or SVG (300 dpi).

- **Ion charge (q)** — formal charge of the ion in units of elementary charge (e.g., 1 for Li^+, 2 for Zn^{2+}, -2 for O^{2-}).
- **Vacancy concentration (%)** — percentage of working-ion sites to be vacated (e.g., 5 = 5%).

- The program automatically reports how many atoms will be removed.

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- **E_m (eV)** — the migration barrier. This can be taken from **DFT-NEB calculations**, bond-valence site energy (BVSE) landscapes, or experimental estimates.
- **Enable ion–ion repulsion** — activates a softBV-style repulsion model [1]:
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If the structure contains a guest water molecule and the diffusing species is a proton - as in MOFs with water in the pores or in hydrated solid polymer electrolytes - the water protons must be separated from the structural hydrogens (C-H, N-H and O-H groups of the framework). FastConduct selects the mobile species by the element symbol only and strips digits from labels (H1, H2 -> H), so the only reliable way to separate the two kinds of hydrogen is a dedicated symbol: Hw.

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1. In the CIF file set _atom_site_type_symbol = Hw for the two hydrogens of every water molecule; structural hydrogens keep the symbol H. The helper script mark_water_h.py does this automatically (it treats any oxygen with two hydrogens within 1.25 A as a water molecule) and can convert the file back before a DFT calculation.
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The results report contains the line 'Species: protons of a guest water molecule (Hw)'. If H is selected while Hw is present, the program warns that structural hydrogens will also be counted.

Important: Hw is not a chemical element. VESTA, ASE and pymatgen will not open such a CIF, and the file must not be used as DFT input. Convert it back with mark_water_h.py --unmark.

4.11. Diffusion Dimensionality

- **1D** — chains / tunnels: $D = \langle MSD \rangle / (2t)$
- **2D** — layers / planes: $D = \langle MSD \rangle / (4t)$
- **3D** — bulk / volume: $D = \langle MSD \rangle / (6t)$

4.12. kMC Simulation Parameters

- **Number of runs** — independent simulation runs (recommended: 5–10) for statistical averaging.
- **Available jumps (N_p)** — number of neighboring sites considered for each ion (auto-detected from the structure).
- **kMC steps per run** — maximum number of Monte Carlo steps (default: 10,000,000).
- **Equilibration steps** — steps performed before data collection to thermalize the system.

- **Output CIF folder** — directory name for saving vacancy structures.

4.13. Temperatures (K)

- Enter up to 5 temperatures separated by commas, e.g.: 250, 275, 300, 325, 350

4.14. Calculation Parameters

- **Attempt frequency ν_0 (Hz)** — typical value: 10^{13} Hz.
- **Max jump length (Å)** — maximum allowed hop distance; 6.5 Å recommended for cations, 4.5 Å for anions.
- **Max simulation time (s)** — physical time limit per temperature.

4.15. Results & Plot

- **Results** text area — full report with conductivity, diffusivity, and statistics. Right-click to copy or save.
- **Arrhenius Plot** — interactive matplotlib figure with linear fit, R^2, and activation energy.
- **Save Plot** — export the figure as PNG, PDF, or SVG (300 dpi).

5. Step-by-Step Workflow

Step 1. Prepare the Input Structure

Prepare a VASP (POSCAR/CONTCAR) or CIF file. Ensure that:

- Coordinates are given in fractional or Cartesian format.

For CIF files, check that the symmetry operations are present (_symmetry_equiv_pos_as_xyz or _space_group_symop_operation_xyz). If only the space-group symbol is given, the program can load just the asymmetric unit and will warn about it.

If the file contains a guest molecule and only its protons should diffuse, label them Hw (see the Hw convention section).

- For CIF files, lattice parameters (_cell_length_a, etc.) are correctly specified.

Step 2. Load the Structure

1. Click **Load File** and select your structure.
2. Verify that the chemical formula and lattice parameters are correct.
3. If loading fails, check the file format.

Step 3. Select the Working Ion

1. Choose the mobile ion from the **Ion** dropdown.
2. Enter the **ion charge** (e.g., 1 for Li^+).
3. Set the **vacancy concentration** (typically 1–10%). The program shows how many atoms will be removed.

Step 4. Set Supercell Dimensions

- Choose **Nx, Ny, Nz** (start with 2×2×2 or 3×3×3).
- Balance between statistical quality and computational cost.

Step 5. Enter Migration Energy

- Input $\mathbf{E_m}$ in eV. This is a key advantage of FastConduct: you can use **DFT-NEB barriers** or any other high-accuracy source rather than relying on fixed empirical potentials.
- If using the repulsion model, enable it and set **alpha** and **cutoff**.

Step 6. Configure kMC Settings

- **Number of runs**: 5–10 for reliable statistics.
- **kMC steps**: 10^7 is usually sufficient; increase for low temperatures.
- **Max simulation time**: 10^{-6} s is a good starting point.

Step 7. Define Temperatures

- Enter 3–5 temperatures covering the operating range of your material (e.g., 250–500 K).

Step 8. Run the Simulation

- Click **RUN kMC SIMULATION**.
- Monitor the progress bar. Runtime depends on the number of runs, temperatures, and supercell size (seconds to minutes).
- You can click **Cancel** at any time to abort. Runtime depends on the number of runs, temperatures, and supercell size (seconds to minutes).

Step 9. Analyze Results

- Review the **Results** panel for conductivity (σ), diffusivity (D), mean hop rates, and MSD at each temperature.
- Examine the **Arrhenius plot**; the slope yields the effective activation energy E_a.

Step 10. Save Outputs

- **Text**: Right-click in the Results area → **Save to file...**
- **Plot**: Click **Save Plot** to export in PNG / PDF / SVG.
- **CIF files**: Automatically saved to the specified output folder.

6. Interpreting the Results

Output Quantities

Quantity	Symbol	Unit	Description
Ionic conductivity	σ	S/cm	From Nernst–Einstein relation
Diffusivity	D	cm^2/s	From mean-squared displacement
Hop rate	rate	Hz	$\nu_0 \cdot \exp(-E_m/kT)$
Standard deviation	±	S/cm	Statistical error across runs
Activation energy (slope)	E_a	eV	Extracted from Arrhenius fit

Arrhenius Plot

- **X-axis**: 1000/T (K^{-1})
- **Y-axis**: $\log_{10}\sigma$ (S/cm)

The effective activation energy is derived from the slope:

$$\mathbf{E_a = -slope \times 2.303 \times k \times 1000/e}$$

Without repulsion, E_a should closely match the input E_m. With repulsion enabled, $E_a > E_m$ due to additional energy penalties from local ion crowding.

7. Troubleshooting

Problem	Cause	Solution
CIF fails to load	File is not a CIF/POSCAR, or the atom loop is malformed	Open the file in VESTA to check it. The v1.2 reader handles GSAS2CIF/FullProf output, several data blocks and Cartesian coordinates
N_p not detected	<2 atoms of selected ion in structure	Check that the working ion exists in the file
σ is extremely low	Too few steps or low temperature	Increase kMC steps or max simulation time
Launch error	Invalid parameter value	Ensure all fields contain valid numbers
Program freezes	Supercell is too large	Reduce Nx, Ny, Nz or increase RAM
CIF files not saved	No write permission	Run from a user-writable directory
Warning: 'Space group found, but symmetry operations could not be parsed'	The CIF gives only the space-group symbol, without an _symmetry_equiv_pos_as_xyz loop	Only the asymmetric unit was loaded. Export a CIF that contains symmetry operations (VESTA: File -> Export Data) or a full P1 structure
Warning: 'The structure may have been parsed incorrectly' (CIF Check)	The atom count does not match _chemical_formula_sum: disordered sites, several data blocks or missing symmetry	Compare the cell and stoichiometry in VESTA; for disordered structures prepare an ordered supercell
All hydrogens are treated as mobile	Water protons and structural hydrogens share the symbol H	Use the Hw convention for the water protons (see the dedicated section)
Conductivity is far too high for a proton conductor	The proton jump network includes framework C-H/N-H groups	Label water protons as Hw, then select Hw as the working ion

8. Theoretical Background

8.1. BKL Rejection-Free kMC Algorithm

FastConduct implements the **rejection-free** (rate-catalog) kMC algorithm following Bortz, Kalos, and Lebowitz [2]:

1. For each active ion, all possible jumps to vacant neighboring sites are identified.
2. Each jump is assigned a rate: $\mathbf{rate = \nu_0 \cdot \exp(-E_{eff} / kT)}$, where $\mathbf{E_eff = E_m + \Delta E_{rep}}$ ($\mathbf{\Delta E_{rep}}$ is used if specified by the user)
3. A jump is selected with probability proportional to $\mathbf{rate_i / \Sigma rate_j}$.
4. The simulation clock advances by $\mathbf{dt = -\ln(r) / \Sigma\, rate_i}$, where $r \in (0,1)$ is a uniform random number.

Note: While a *rejection-based* (null-event) kMC variant exists in the literature [3], FastConduct uses the rejection-free approach because it is computationally more efficient when the number of possible transitions is large.

9.2. Ion–Ion Repulsion Model (softBV-Style)

When **Enable ion–ion repulsion** is active, the program evaluates local repulsion analogously to the softBV method [1]:

$$E_{rep} = \alpha \cdot N_{occupied\ neighbors}$$

$$\Delta E = E_{rep(target)} - E_{rep(start)}$$

Here, α is the repulsion energy per occupied neighbor, and $N_{occupied\ neighbors}$ counts occupied sites within the specified cutoff. This accounts for the effect of local ion crowding on the migration barrier.

9.3. Nernst–Einstein Equation

Conductivity is computed via the Nernst–Einstein relation [2]:

$$\sigma = n \cdot q^2 \cdot D/(k \cdot T)$$

where:

- **n** — carrier concentration (cm^{-3})
- **q** — ionic charge (C)
- **D** —diffusivity (cm^2/s)
- **T** — absolute temperature (K)

Note: Some formulations include the Haven ratio H_r in the denominator. FastConduct assumes $\mathbf{H_r = 1}$, which corresponds to the ideal dilute-defect limit where successive hops are uncorrelated.

9.4. Diffusivity and MSD

The mean-squared displacement (MSD) is accumulated with full periodic-boundary correction. The diffusion coefficient depends on dimensionality [4]:

- **1D:** $D = \langle MSD \rangle/(2t)$
- **2D:** $D = \langle MSD \rangle/(4t)$
- **3D:** $D = \langle MSD \rangle/(6t)$

10. Citation & References

If you use FastConduct in your research, please cite:

FastConduct v1.2 — Ionic Conductivity Calculator (kMC)

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11. Acknowledgments

FastConduct v1.2 was developed using the following open-source libraries and tools:

- **Python 3.13+** — core programming language
- **NumPy** — numerical computations and array operations
- **Matplotlib** — generation of Arrhenius plots and scientific figures

- **SciPy** — scientific computing (cKDTree for neighbor search, physical constants)
- **PIL / Pillow** — image handling for the GUI logo and splash screen
- **CustomTkinter — modern graphical user interface framework**

The author gratefully acknowledges the developers and maintainers of these projects for providing the robust scientific software ecosystem that made this application possible.

Methodological References

[1] H. Chen, L.L. Wong, S. Adams, SoftBV – a software tool for screening the materials genome of inorganic fast ion conductors, Acta Crystallogr. Sect. B Struct. Sci. Cryst. Eng. Mater. 75 (2019) 18–33. https://doi.org/10.1107/S2052520618015718.

[2] A.B. Bortz, M.H. Kalos, J.L. Lebowitz, A new algorithm for Monte Carlo simulation of Ising spin systems, J. Comput. Phys. 17 (1975) 10–18. https://doi.org/10.1016/0021-9991(75)90060-1.

[3] S. Eisele, S. Grieshammer, MOCASSIN: Metropolis and kinetic Monte Carlo for solid electrolytes, J. Comput. Chem. 41 (2020) 2663–2677. https://doi.org/https://doi.org/10.1002/jcc.26418.

[4] P. Hein, B.O.H. Grope, J. Koettgen, S. Grieshammer, M. Martin, Kinetic Monte Carlo simulations of ionic conductivity in oxygen ion conductors, Mater. Chem. Phys. 257 (2021) 123767. https://doi.org/10.1016/j.matchemphys.2020.123767.

Successful simulating!

For questions or feature requests, please contact the author via ResearchGate, ORCID or open an issue on GitHub